DSEH001 · DIGITAL TRANSFORMATION IN HR SYSTEMS

03

UNIT 3 READER

# Analytics, Algorithms and Accountability

*People analytics, fairness audits and data storytelling*

Weeks 7 to 9 | Round 4: The Audit | Round 5: The Story

15 readings explained

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# About this reader

**How to use it.** This reader is a companion to the set readings, not a replacement for them. For each reading it explains the argument in plain language, lists the ideas you must hold, links them to GreenLeaf and the Disha rounds, flags common confusions, and gives questions to think with. Read the original first, then use this reader to check your understanding. Folder names refer to the course Readings folder.

Unit 3 is about what happens when HR decisions start to depend on data and algorithms. The first readings show how analytics can move from describing the past to predicting the future. The governance readings then ask how an organization should test whether a model is accurate, fair and controlled before it is allowed to shape people's careers. The final readings turn to communication: how to tell an honest story with data, and how algorithms change the balance of power between employers and workers.

Hold two ideas together as you read. First, a model is a claim, not a fact; it must be audited against what actually happened. Second, fairness is judged both by outcomes, such as selection rates, and by process, such as whether people have a voice and a route to appeal. Rounds 4 and 5 test both ideas on GreenLeaf's recruitment and flight risk data.

### READING MAP

| Week | Set readings | Folder | Disha link |
|---|---|---|---|
| 7 | Marler and Boudreau (2017); Davenport, Harris and Shapiro (2010); Fitz-enz and Mattox (2014) Ch 1 and 2. Round 4 frameworks: NIST AI RMF 1.0; ISO/IEC 23894; EEOC four fifths rule | Unit-2 Supp., Unit-3 Core, Unit-3 Supp. | Round 4 |
| 8 | Kavanagh and Johnson (2018) Ch 14; Angrave et al. (2016); an Indian HR dashboard case. Round 5 framework: Knaflic (2015) | Unit-2 Core, Unit-3 Core, Unit-3 Supp. | Round 5 |
| 9 | Tambe et al. (2019); Kellogg et al. (2020); Leicht-Deobald et al. (2019); DPDP employee data note | Unit-3 Core, Unit-4 Supp. | Round 4 debrief; Round 5 briefing |

# The readings, one by one

Week 7 | Core

Marler, J. H., & Boudreau, J. W. (2017). An evidence based review of HR analytics. *The International Journal of Human Resource Management*, 28(1), 3 to 26.

### THE ARGUMENT IN BRIEF

Marler and Boudreau set out to answer a simple question: what does the published evidence actually say about HR analytics? They define HR analytics as an HR practice enabled by information technology that uses descriptive, visual and statistical analysis of HR processes, human capital and organizational performance to establish business impact and enable data driven decisions. Searching the research literature, they found relatively few peer reviewed studies, and much of what existed was practitioner writing. Their review summarizes what is known about how HR analytics is defined, what helps or hinders its adoption, and whether it improves organizational outcomes. Their conclusion is cautious: the promise is widely asserted, but the evidence base is still thin.

### IDEAS TO HOLD

- A working definition: analytics that links HR data to business impact and decisions.
- Peer reviewed evidence on HR analytics was limited at the time of the review.
- Adoption depends on analytical skill, data quality and the ability to tell a business story.
- Evidence that analytics improves performance is promising but not yet strong.

**CONNECT TO GREENLEAF**

Use this as your benchmark in Round 4. Before claiming that GreenLeaf's analytics function will 'improve retention', ask what evidence would show it, and whether the flight risk model meets the definition at all.

**EASY TO CONFUSE**

This is a review of evidence about HR analytics. Angrave et al. (2016) is an argument about why HR analytics fails. They agree on caution but do different work.

**QUESTIONS TO THINK WITH**

- Does GreenLeaf's flight risk model meet Marler and Boudreau's definition of HR analytics? Which part is missing?

Week 7 Core

Davenport, T. H., Harris, J., & Shapiro, J. (2010). Competing on talent analytics. *Harvard Business Review*, 88(10), 52 to 58.

### THE ARGUMENT IN BRIEF

This practitioner article argues that leading firms gain advantage by applying analytics to their people, not only to customers and operations. Drawing on company examples, the authors describe a progression of talent analytics, from basic facts about the workforce, through analysis of which HR practices drive performance and forecasts of future workforce needs, to models of what employees value and how talent flows through the organization. They stress that analytics needs leadership sponsorship and a culture that will act on evidence.

### IDEAS TO HOLD

- Talent analytics ranges from basic workforce facts to forecasting and value modeling.
- Its value comes from linking people data to business performance.
- Sponsorship and a culture of acting on evidence matter as much as tools.

#### CONNECT TO GREENLEAF

Place each analytics question your pod asked in Round 4 on Davenport and colleagues' progression. Which ones are only facts, and which ones forecast?

#### QUESTIONS TO THINK WITH

- Which step of the progression would GreenLeaf reach first, and what would it need to climb one step higher?

Week 7 Core

Fitz-enz, J., & Mattox, J. R., II. (2014). *Predictive Analytics for Human Resources*. Wiley. Chapter 1 (Where's the Value?) and Chapter 2 (Getting Started).

### THE ARGUMENT IN BRIEF

Fitz-enz and Mattox argue that HR analytics should move beyond reporting what happened toward anticipating what will happen and guiding action. The opening chapters set out the case for predictive work and the conditions it needs: a clear business question, data aligned to that question, and a culture that will act on the results. They stress that

prediction is valuable only when it connects people measures to outcomes the business cares about, and that the analysis must lead to a decision someone owns.

#### IDEAS TO HOLD

- Analytics progresses from describing, to explaining, to predicting.
- Every model starts with a business question, not with the data available.
- Prediction matters only if it leads to an owned decision.
- Data quality and culture limit what analytics can achieve.

> **CONNECT TO GREENLEAF**
>
> Before judging GreenLeaf's flight risk model, write the business question it was meant to answer and the decision it should inform.

> **QUESTIONS TO THINK WITH**
>
> - What decision would GreenLeaf take differently if the flight risk model were perfectly accurate?

**Week 7** **Round 4 framework**

National Institute of Standards and Technology. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)* (NIST AI 100 1).

#### THE ARGUMENT IN BRIEF

The NIST framework gives organizations a voluntary structure for managing AI risk through four functions. Govern sets the culture, policies, roles and accountability for AI risk and runs across all the others. Map establishes the context: the purpose of the system, who is affected and what could go wrong. Measure uses methods and metrics to assess risks, including tests of accuracy and bias. Manage allocates resources to treat the risks found, including deciding whether to continue, change or stop a system. The framework names characteristics of trustworthy AI, including validity and reliability, safety, security, accountability and transparency, explainability, privacy, and fairness with harmful bias managed.

#### IDEAS TO HOLD

- Four functions: Govern, Map, Measure, Manage.
- Govern is cross cutting; it assigns accountability for the rest.
- An audit of model accuracy or bias is a Measure activity.
- Deciding to pause or retire a system is a Manage activity.

**CONNECT TO GREENLEAF**

Your Round 4 audit is Measure work. Your recommendation about what GreenLeaf should do next belongs to Manage, and who signs it off belongs to Govern.

**EASY TO CONFUSE**

Do not confuse the NIST functions with ISO/IEC 23894. NIST organizes the whole risk program; the human review specification describes one control within it.

**QUESTIONS TO THINK WITH**

- A team computes the false positive rate of the flight risk model. Which NIST function is this? Who should own the response?

**Week 7** **Round 4 framework**

ISO/IEC 23894:2023. *Information technology: Artificial intelligence: Guidance on risk management*. ISO.

**THE ARGUMENT IN BRIEF**

ISO/IEC 23894 applies the general risk management approach of ISO 31000 to AI. It guides organizations to identify, analyze, evaluate and treat AI specific risks across the system's life cycle, and to integrate this work into normal management. Human oversight is one of the treatments it supports. In this course, Round 4 asks you to write a human review specification that answers four questions: who reviews, how often, against what criteria, and what the escalation path is. This four part specification is the course's own and should be credited to the course.

**IDEAS TO HOLD**

- AI risk management follows the identify, analyze, evaluate, treat cycle.
- Risk work continues across the whole life cycle, not only at launch.
- Human oversight is a treatment that must be specified to be real.
- The course's specification: who, cadence, criteria, escalation.

**CONNECT TO GREENLEAF**

A sentence such as 'HR will review the model's outputs' is not a specification. Name the role, the frequency, the threshold that triggers review, and where disputes go.

**EASY TO CONFUSE**

Human review corrects decisions; explainability describes them. An explanation dashboard is not a human review process.

**QUESTIONS TO THINK WITH**

- Write the four parts of a human review specification for GreenLeaf's screening tool.

**Week 7** **Round 4 framework**

U.S. Equal Employment Opportunity Commission et al. (1978). *Uniform Guidelines on Employee Selection Procedures*, 29 CFR 1607.4(D).

**THE ARGUMENT IN BRIEF**

The Uniform Guidelines set a practical rule of thumb for adverse impact. Compute the selection rate for each group: the number selected divided by the number who applied from that group. If the rate for one group is less than four fifths (80 percent) of the rate for the group with the highest rate, this is generally regarded as evidence of adverse impact. The rule looks at outcomes, not intent, so a process can show adverse impact even if no one meant to discriminate. It is a screening test that prompts investigation, not final proof of discrimination, and small samples call for caution.

**IDEAS TO HOLD**

- Selection rate = selected ÷ applicants, computed for each group.
- Impact ratio = lower group's rate ÷ highest group's rate; below 0.80 flags adverse impact.
- Intent is irrelevant; the test is about outcomes.
- A flag starts an investigation; it does not end one.

**CONNECT TO GREENLEAF**

In Recruitment_ATS, compare rates, not counts: offers to women divided by female applications, against the male equivalent. Split by Algorithm_Used before drawing a conclusion.

**EASY TO CONFUSE**

Comparing raw offer counts is the most common error. More offers to men can be fair if more men applied; the test uses rates.

**QUESTIONS TO THINK WITH**

- Why can raw counts and rates point in opposite directions? Construct a small example.

**Week 7** **Round 4 framework**

Explainable AI and the precautionary principle.

## THE ARGUMENT IN BRIEF

Explainable AI methods show which inputs a model relied on, which helps people understand and contest its outputs. But an explanation is not a correction: showing that a model is biased does not make it unbiased. The precautionary principle, set out in Principle 15 of the 1992 Rio Declaration, holds that lack of full certainty should not delay action against serious harm. Applied to HR algorithms, when harm to people is evident and benefit is unproven, the burden of proof falls on those who want to continue using the system.

## IDEAS TO HOLD

- Explanation makes a model's reasoning visible; it does not fix it.
- An explainability dashboard is not a remedy for bias.
- Precaution: with evident harm and unproven benefit, the case must be made for continuing.
- Precaution is a decision rule; NIST's Manage function is where it is applied.

**CONNECT TO GREENLEAF**

If your audit finds adverse impact and the tool's benefit is unproven, ask who must prove what before the tool is used again.

**QUESTIONS TO THINK WITH**

- When would a precautionary pause be disproportionate? What evidence would justify continuing?

Week 8 Core

Kavanagh, M. J., & Johnson, R. D. (2018). Chapter 14 (dashboards).

### THE ARGUMENT IN BRIEF

Read Chapter 14 again this week with dashboards in mind. A dashboard should present a small number of well defined metrics tied to decisions, with targets and context, for a named audience. Overloaded dashboards that display every available number hide the signal. The authors also note that metrics can drive behavior, so the choice of what to show is itself a management decision.

### IDEAS TO HOLD

- Fewer, decision linked metrics beat many unexplained ones.
- Each metric needs a target, a trend and an owner.
- What you display shapes what people manage.

### CONNECT TO GREENLEAF

Design your Round 5 dashboard for the Board, not for analysts. Every tile should answer a question the Board will actually ask.

### QUESTIONS TO THINK WITH

- Which three GreenLeaf metrics would you keep if a Board member allowed only three?

Week 8 Core

Angrave, D., et al. (2016). HR and analytics: Why HR is set to fail the big data challenge. *Human Resource Management Journal*, 26(1), 1 to 11.

### HOW TO READ IT

First set in Week 3 and explained in the Unit 1 Reader. Read it again now with dashboards and KPIs in mind: Angrave and colleagues warn that tools without analytical skill and theory produce counts of what is easy to count. Ask of every tile on your Round 5 dashboard whether it reflects a real question or only available data.

Week 8 Case

An Indian HR dashboard or data culture case (shared in class)

### HOW TO READ IT

Read the case for two things: which decision each dashboard element was meant to drive, and what the organization did to build data literacy among the people who use it. Compare both with your pod's Round 5 KPI framework and data culture plan.

Week 8 Round 5 framework

Knaflic, C. N. (2015). *Storytelling with Data: A Data Visualization Guide for Business Professionals*. Wiley.

### THE ARGUMENT IN BRIEF

Knaflic teaches that effective data communication starts with the audience and the action you want them to take. Her lessons move in sequence: understand the context, choose an appropriate visual, remove clutter, focus the audience's attention, think like a designer, and tell a story. She recommends stating the main idea as one complete sentence before any chart is built, and testing whether the whole story can be told briefly without slides. Decluttering is not decoration; every element that does not help the message competes with it.

### IDEAS TO HOLD

- Start with who the audience is and what you want them to do.
- Write the headline as a full sentence that states the finding.
- Declutter: remove what does not serve the message.
- Use color and position to direct attention to the one thing that matters.

### CONNECT TO GREENLEAF

A Round 5 headline such as 'Attrition overview' is a label. 'The model wrongly flags many staff who stay' is a headline. Pair each headline with one action and a counter metric that would show if the action backfires.

### EASY TO CONFUSE

A dashboard that shows everything is a data dump, not a story.

**QUESTIONS TO THINK WITH**

- Rewrite one of your Round 5 chart titles as a complete sentence.

**Week 9** **Core**

Tambe, P., Cappelli, P., & Yakubovich, V. (2019). Artificial intelligence in human resources management: Challenges and a path forward. *California Management Review*, 61(4), 15 to 42.

**THE ARGUMENT IN BRIEF**

Tambe and colleagues identify four challenges that make AI harder to apply in HR than in marketing or finance. HR outcomes are complex and hard to measure; good performance is not a single number. Many HR datasets are small, because organizations hire and lose relatively few people. Fairness and legal constraints apply to decisions about people. And employees react to being managed by algorithms, which can change their behavior. The authors propose a path forward that includes using causal reasoning, accepting randomness where outcomes cannot be predicted, and involving employees in the design of systems.

**IDEAS TO HOLD**

- HR outcomes are complex and often poorly measured.
- Small data limit what machine learning can reliably learn.
- Fairness and legal accountability constrain algorithmic decisions about people.
- Employee reactions can undermine a technically sound system.

**CONNECT TO GREENLEAF**

GreenLeaf's 1,000 row sample is a small data problem. Be cautious about any model claim that rests on a handful of exits in one segment.

**QUESTIONS TO THINK WITH**

- Which of the four challenges is most serious for GreenLeaf's flight risk model? Defend your choice.

Week 9 Core

Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals*, 14(1), 366 to 410.

### THE ARGUMENT IN BRIEF

Kellogg and colleagues review how algorithms are used to control workers. They describe six mechanisms: algorithms can restrict and recommend (directing what workers do), record and rate (evaluating them), and replace and reward (disciplining them). Compared with earlier forms of control, algorithmic control is more comprehensive, more immediate and less transparent. The authors also describe how workers resist, which they call algoactivism: gaming metrics, coordinating with peers, and pressing for change. Their conclusion is that the workplace becomes a new contested terrain where power is negotiated through technology.

### IDEAS TO HOLD

- Six mechanisms: restrict, recommend, record, rate, replace, reward.
- Algorithmic control is more comprehensive, instant and opaque.
- Workers respond with algoactivism: individual and collective resistance.
- Control and resistance are shaped by how transparent the system is.

> **CONNECT TO GREENLEAF**
>
> Gig_Workforce's Algorithmic_Allocation_Score uses weights that are not disclosed to partners. Map which of the six mechanisms that algorithm performs.

> **QUESTIONS TO THINK WITH**
>
> - Which mechanisms are visible in GreenLeaf's gig allocation, and what form of resistance might they provoke?

Week 9 Statute

The DPDP employee data note (on the Digital Personal Data Protection Act, 2023)

### HOW TO READ IT

The Act itself is explained in the Unit 1 Reader. For Week 9, focus on employee data: Section 7 lets an employer process personal data without consent for purposes connected with employment, but processing must be necessary and limited to those purposes. Ask

whether a flight risk model or a gig allocation algorithm fits within that limit, or needs another basis.

Week 9 Core

Leicht-Deobald, U., Busch, T., Schank, C., Weibel, A., Schafheitle, S., Wildhaber, I., & Kasper, G. (2019). The challenges of algorithm based HR decision making for personal integrity. *Journal of Business Ethics*, 160(2), 377 to 392.

### THE ARGUMENT IN BRIEF

This article argues that algorithm based HR decisions can erode personal integrity, meaning people's capacity to act on their own values. When decisions are delegated to systems that cannot be questioned, managers may follow recommendations they do not understand, and employees may adjust their behavior to satisfy the system rather than to do good work. The authors call for remedies that keep human judgment alive, including critical data literacy, ethical awareness, participatory design that involves employees, and forms of regulation.

### IDEAS TO HOLD

- Algorithmic decisions can crowd out moral judgment.
- Blind compliance by managers is a risk, not only bias in the model.
- Remedies include critical data literacy and employee participation in design.

> **CONNECT TO GREENLEAF**
>
> Use this reading to argue why GreenLeaf needs human review that can actually overrule the model, not only approve it.

> **QUESTIONS TO THINK WITH**
>
> - What would 'critical data literacy' look like for a GreenLeaf store manager reading a flight risk score?

## Audit and story: the concepts students most often mix up

| Concept | What it does | Not the same as |
|---|---|---|
| Four fifths rule | Compares selection rates between groups; below 0.80 flags impact | Comparing raw offer counts |
| False positive rate | False positives ÷ (false positives + true negatives): among those who stayed, the share flagged | False positives ÷ all employees |
| Human review (course specification) | Who, cadence, criteria, escalation; can change decisions | Explainability |
| Explainable AI | Shows what the model relied on | A fix for bias |
| NIST Measure | Testing accuracy and bias | NIST Manage (acting on results) |
| Spurious correlation and proxies | A heavily weighted variable may stand in for another cause | Causal effect size |
| Procedural justice | Fairness of process: voice, consistent rules, correctability | A fast grievance SLA |

### THREE ROUND 5 IDEAS TO CARRY FORWARD

**Correlation, proxies and the confusion matrix.** A model may weight a variable heavily because it is correlated with leaving, not because it causes leaving; it may also be a proxy for something else, such as grade or location. Test such claims by comparing outcomes within bands of the variable. To evaluate the model, build a confusion matrix from Predicted_Attrition against Actual_Attrition in PeopleAnalytics_Fact.

**Procedural justice.** Research since Thibaut and Walker (1975) and Leventhal (1980) shows that people judge fairness by process as well as outcome: whether they had a voice, whether rules were applied consistently, whether decisions can be corrected. A fast grievance process is welcome, but speed is not proof of fair outcomes.

**Crisis communication and alternative dispute resolution.** Good crisis communication buys time and trust; it does not resolve the underlying problem. Alternative dispute resolution, such as an independent ombudsperson, gains credibility from independence, and the price of that credibility is that management gives up some control over the result.

# Key terms

| | |
|---|---|
| **Adverse impact** | A substantially lower selection rate for one group, regardless of intent. |
| **Algoactivism** | Kellogg et al.'s term for worker resistance to algorithmic control. |
| **Confusion matrix** | A table of true positives, false positives, true negatives and false negatives. |
| **Counter metric** | A measure that would reveal if an action creates harm elsewhere. |
| **Proxy** | A variable that stands in for another, often hidden, characteristic. |
| **Selection rate** | Number selected divided by number of applicants in a group. |

# References


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*The four part human review specification (who, cadence, criteria, escalation) is the course's own and is not credited to ISO/IEC 23894.*